

Performance of GPU-Accelerated VASP 6.5.1 on a Dual NVIDIA H100 Node:

Materials Project PBE and r²SCAN Workflows, and Finite-Displacement Phonons

Gilles Frapper
Professor, Theoretical Chemistry
IC2MP UMR 7285 CNRS, Université de Poitiers
gilles.frapper@univ-poitiers.fr

Draft, September 27, 2026

Abstract

[TODO: Write once all results are final (2-GPU series and phonons).] We benchmark the GPU (OpenACC) build of VASP 6.5.1 on a shared node with two NVIDIA H100 NVL GPUs. Single-point and full cell relaxation calculations are run on supercell series of Si, Al, and MgO from 1 to 64 atoms, using the Materials Project PBE and r²SCAN input sets. We report wall time, time per self-consistent step, GPU memory, the cost of r²SCAN relative to PBE, the speedup from one to two GPUs with two parallelization strategies, run-to-run reproducibility, and agreement with published Materials Project energies. Finite-displacement phonon calculations complete the study.

1 Introduction

[TODO: Context: GPU ports of plane-wave DFT codes; high-throughput workflows such as Materials Project [1]; practical questions for a small shared GPU node.]

This study addresses the following questions:

1. How does computing time scale with the number of atoms, from 1 to 64?
2. How much more does r²SCAN cost than PBE, per self-consistent (SCF) step and in total?
3. How much does a full relaxation cost compared with a single point?
4. What is gained by going from one to two GPUs, and is it better to distribute k -points (KPAR=2) or bands and plane waves (KPAR=1)?
5. How much GPU memory is needed as a function of system size?
6. Are timings reproducible from one job to the next?
7. Are the computed energies consistent with those published by Materials Project?
8. What is the time-to-solution of a complete finite-displacement phonon calculation on one and two GPUs?

2 Hardware and Software

Table 1: Test system.

Component	Specification
GPUs	$2 \times$ NVIDIA H100 NVL, 94 GB each (driver 595.84)
CPU	AMD EPYC 7662, 32 cores assigned to the virtual machine
Host memory	62 GB
Per GPU (Slurm)	16 cores, $\sim$ 28 GB host memory, no time limit
DFT code	VASP 6.5.1, OpenACC GPU build
Toolchain	NVIDIA HPC SDK 26.3, HPC-X (CUDA-aware Open MPI), Intel oneMKL
Execution	one MPI rank per GPU, <code>OMP_NUM_THREADS=1</code>
Workflow tools	pymatgen 2026.9.23 [2], phonopy 4.6.0 [3, 4]

The node is shared with other users through Slurm. To leave the second GPU available, study jobs were submitted as job arrays running one calculation at a time. One-GPU and two-GPU series were never submitted concurrently. For each calculation, the jobs running on the node at start time were recorded, since they may affect timings (Section 3.4).

3 Methods

3.1 Structures

All structures were retrieved from Materials Project [1] (Table 2). To isolate the effect of system size at constant chemistry, each material is studied as a series of supercells of increasing size, 14 structures in total. The MP structures are already relaxed with PBE.

Table 2: Scaling series.

Material	MP ID	Type	Sizes (atoms)
Si, diamond	mp-149	semiconductor (PBE gap 0.61 eV)	2 (primitive), 8, 16, 32, 64
Al, fcc	mp-134	metal	1 (primitive), 4, 32, 64
MgO, rock salt	mp-1265	insulator (PBE gap 4.43 eV)	2 (primitive), 8, 16, 32, 64

3.2 Electronic-structure settings

All calculations use the projector augmented-wave (PAW) method [5, 6] as implemented in VASP [7]. Inputs were generated with the Materials Project input sets implemented in pymatgen [2], without changing any physical parameter (Table 3). The PBE sets [8] apply the MP Hubbard U values [9] to oxides and fluorides of Co, Cr, Fe, Mn, Mo, Ni, and V (none of the present materials is affected). The r^2 SCAN sets [10] follow the MP meta-GGA workflow [11], in which the k -point spacing is chosen from the MP band gap. All calculations are spin-polarized (`ISPIN=2`), with `PREC=Accurate` and `LASPH=.TRUE.`

The following deviations from the MP methodology apply identically to every calculation:

- **Pseudopotentials.** The PBE_54 PAW set installed on the machine is used for both functionals. MP uses an older PBE set for its GGA calculations, which is identical for almost all elements; `W_pv`, absent from PBE_54, would be replaced by `W_sv`.

Table 3: Calculation series and main parameters. Brillouin-zone sampling is expressed per reciprocal-lattice volume (PBE) or as KSPACING (r²SCAN).

Series	pymatgen set	Main parameters
pbe_sp	MPStaticSet	PBE; ENCUT=520 eV; 100 k -points per $\text{\AA}^{-3}$; EDIFF= 5×10^{-5} eV per atom; ALGO=Fast; LREAL=FALSE.
pbe_relax	MPRelaxSet	as above, with ISIF=3, IBRION=2; 64 k -points per $\text{\AA}^{-3}$; LREAL=Auto
r2scan_sp	MPScanStaticSet	METAGGA=R2SCAN; ENCUT=680 eV; KSPACING from 0.22 to $0.44 \text{\AA}^{-1}$ depending on the MP gap; EDIFF= 1×10^{-5} eV
r2scan_relax	MPScanRelaxSet	as above, with ISIF=3, ALGO=All, EDIFFG= $-0.02 \text{ eV/\AA}$

- **Output.** Writing of WAVECAR, CHGCAR, AECCAR, LOCPOT, and ELFCAR is disabled, so that the benchmark measures computation rather than I/O.
- **Parallelization tags.** NCORE is removed, since the GPU build forces it to 1; KPAR is set by the parallelization variant (Section 3.3).
- **Small k -point meshes.** When a mesh has fewer than 4 irreducible k -points, the tetrahedron method (ISMear=-5) is replaced by Gaussian smearing (ISMear=0, SIGMA=0.05 eV), which is also the correction applied by custodian in MP workflows.

In the supercells, the k -point density per atom is kept constant, as in the MP methodology, so the number of k -points decreases as the cell grows. Relaxations start from the MP structures, which are already relaxed with PBE: PBE relaxations therefore converge in a few ionic steps, and r²SCAN relaxations need a few more.

3.3 Parallelization variants

Three variants are compared (Table 4), with identical physical inputs. With two GPUs, KPAR=1 distributes bands and plane-wave coefficients across both devices, while KPAR=2 assigns one group of k -points to each GPU. The g2k2 variant is skipped when a structure has fewer than two irreducible k -points.

Table 4: Parallelization variants.			
Variant	GPUs	MPI ranks	KPAR
g1	1	1	1
g2k1	2	2	1
g2k2	2	2	2

3.4 Measurement protocol

Each calculation runs as one Slurm job-array task launched with `mpirun -np  $N_{\text{GPU}}$  -bind-to none vasp_std`. The following quantities are recorded:

- **Problem size:** number of atoms, NKPTS, NBANDS, NPLWV, NELECT, ISPIN.

- **Time:** total time (“Elapsed time” in OUTCAR, and job wall time); median time per SCF step (LOOP lines); mean time per ionic step (LOOP+ lines); numbers of SCF and ionic steps.
- **Memory and utilization:** peak GPU memory and mean GPU utilization, sampled every 2s with nvidia-smi; peak host memory.
- **Energy:** energy per atom and its difference from the uncorrected MP energy of the same functional, in meV/atom.
- **Context:** other jobs running on the node at start time.

The speedup of a two-GPU variant is the ratio of the one-GPU total time to the two-GPU total time. Reproducibility was assessed by repeating six single-point calculations (Si 8 atoms, Al 32 atoms, and MgO 32 atoms, with PBE and r²SCAN) three times each on one GPU.

The study was carried out in phases, each submitted separately from the smallest to the largest system (Table 5).

Table 5: Study phases.

Phase	Content	GPUs	Calculations
0	Workflow validation on 1–2 atoms	1	5
1	Reproducibility, 3 repeats	1	18
2	Single points, PBE and r ² SCAN	1	28
3	Relaxations, PBE and r ² SCAN	1	28
4	All four series, g2k1 and g2k2	2	112
5	Finite-displacement phonons, g1 and g2k1	1, 2	6 workflows

3.5 Finite-displacement phonons

Harmonic phonons of the primitive cells of Si, Al, and MgO are computed with the finite-displacement method using phonopy [3, 4]. Each material is run as one automated workflow per Slurm task, with identical inputs on one GPU (g1) and two GPUs (g2k1, KPAR=1), so that only the hardware changes.

Settings. PBE with PBE_54 PAW potentials, ENCUT=520 eV, PREC=Accurate, EDIFF=1 × 10^{−8} eV, ADDGRID=.TRUE., LREAL=.FALSE., LASPH=.TRUE., ALGO=Normal, non-spin-polarized. Gaussian smearing with $\sigma = 0.01$ eV is used for Si and MgO, and first-order Methfessel–Paxton smearing with $\sigma = 0.2$ eV for Al. Γ -centered k -meshes are built with $N_i = \max(1, \text{round}(L |\mathbf{b}_i|))$, where $\mathbf{b}_i$ are the reciprocal lattice vectors (without the 2π factor) and $L = 50$ Å for insulators and 80 Å for the metal.

Workflow.

1. **Tight relaxation** of the primitive cell (ISIF=3, EDIFFG=−1 × 10^{−3} eV/Å). The relaxation is restarted from the previous CONTCAR, which resets the plane-wave basis, until a pass completes in at most two ionic steps (at most four passes).
2. **Born effective charges and ε_∞** for polar materials (MgO) from density-functional perturbation theory (LEPSILON), with a finite-field fallback (LCALCEPS). They are used for the non-analytical term correction.

3. **Force calculations** for the symmetry-irreducible displacements ($0.01 \text{ \AA}$) in two diagonal supercell sizes, to check convergence (Table 6).
4. **Post-processing:** symmetrized force constants, band structure along the seekpath path [12], density of states on a 30^3 q -mesh, and thermodynamic properties from 0 to 1,000 K.

Table 6: Phonon supercells.

Material	Relax. k -mesh	Supercells (atoms)	Displacements	Supercell k -meshes
Si	16^3	3^3 (54), 4^3 (128)	1 per cell	5^3 , 4^3
Al	34^3	4^3 (64), 5^3 (125)	1 per cell	9^3 , 7^3
MgO	21^3	3^3 (54), 4^3 (128)	2 per cell	7^3 , 5^3

Metrics. VASP wall time of every step and GPU-hours; time-to-solution from submission to result, queue time included; quality checks: acoustic frequencies at Γ , lowest frequency on the q -mesh (imaginary modes), and residual force drift.

4 Results

4.1 Scaling with system size on one GPU

[TODO: Table of the 56 one-GPU calculations; Fig. t_total_vs_natoms and t_scf_vs_natoms; scaling exponents.]

4.2 Cost of r²SCAN relative to PBE

[TODO: Ratios per SCF step and in total, single points and relaxations.]

4.3 Relaxation versus single point

[TODO:]

4.4 GPU memory

[TODO:]

4.5 Reproducibility

[TODO: Relative spread of the three repeats.]

4.6 Two GPUs: speedup and KPAR

[TODO: Final table and Fig. speedup_2gpu once phase 4 is complete.]

4.7 Agreement with Materials Project energies

[TODO: Energy differences per series; explain the PBE offsets for MgO and Al.]

4.8 Phonons

[TODO: Dispersions, convergence between supercell sizes, time-to-solution on one and two GPUs.]

5 Discussion and Recommendations

[TODO: Practical guidance for users of the node: when to request two GPUs, choice of KPAR, memory headroom, expected run times.]

6 Conclusion

[TODO:]

Data and Code Availability

Scripts, inputs generation, and result tables are available at https://github.com/John-Akapulco/GPU_vmp_testing_VASP_etc. VASP output files are kept on the test machine; PAW potentials are not redistributed, in accordance with the VASP license.

References

- [1] A. Jain, S. P. Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, and K. A. Persson. Commentary: The Materials Project: A materials genome approach to accelerating materials innovation. *APL Mater.*, 1:011002, 2013. doi: 10.1063/1.4812323.
- [2] S. P. Ong, W. D. Richards, A. Jain, G. Hautier, M. Kocher, S. Cholia, D. Gunter, V. L. Chevrier, K. A. Persson, and G. Ceder. Python materials genomics (pymatgen): A robust, open-source python library for materials analysis. *Comput. Mater. Sci.*, 68:314–319, 2013. doi: 10.1016/j.commatsci.2012.10.028.
- [3] A. Togo, L. Chaput, T. Tadano, and I. Tanaka. Implementation strategies in phonopy and phono3py. *J. Phys.: Condens. Matter*, 35:353001, 2023. doi: 10.1088/1361-648X/acd831.
- [4] A. Togo and I. Tanaka. First principles phonon calculations in materials science. *Scr. Mater.*, 108:1–5, 2015. doi: 10.1016/j.scriptamat.2015.07.021.
- [5] P. E. Blöchl. Projector augmented-wave method. *Phys. Rev. B*, 50:17953–17979, 1994. doi: 10.1103/PhysRevB.50.17953.
- [6] G. Kresse and D. Joubert. From ultrasoft pseudopotentials to the projector augmented-wave method. *Phys. Rev. B*, 59:1758–1775, 1999. doi: 10.1103/PhysRevB.59.1758.
- [7] G. Kresse and J. Furthmüller. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B*, 54:11169–11186, 1996. doi: 10.1103/PhysRevB.54.11169.
- [8] J. P. Perdew, K. Burke, and M. Ernzerhof. Generalized gradient approximation made simple. *Phys. Rev. Lett.*, 77:3865–3868, 1996. doi: 10.1103/PhysRevLett.77.3865.

- [9] S. L. Dudarev, G. A. Botton, S. Y. Savrasov, C. J. Humphreys, and A. P. Sutton. Electron-energy-loss spectra and the structural stability of nickel oxide: An LSDA+U study. *Phys. Rev. B*, 57:1505–1509, 1998. doi: 10.1103/PhysRevB.57.1505.
- [10] J. W. Furness, A. D. Kaplan, J. Ning, J. P. Perdew, and J. Sun. Accurate and numerically efficient r²SCAN meta-generalized gradient approximation. *J. Phys. Chem. Lett.*, 11:8208–8215, 2020. doi: 10.1021/acs.jpclett.0c02405.
- [11] R. Kingsbury, A. S. Gupta, C. J. Bartel, J. M. Munro, S. Dwaraknath, M. Horton, and K. A. Persson. Performance comparison of r²SCAN and SCAN metaGGA density functionals for solid materials via an automated, high-throughput computational workflow. *Phys. Rev. Mater.*, 6:013801, 2022. doi: 10.1103/PhysRevMaterials.6.013801.
- [12] Y. Hinuma, G. Pizzi, Y. Kumagai, F. Oba, and I. Tanaka. Band structure diagram paths based on crystallography. *Comput. Mater. Sci.*, 128:140–184, 2017. doi: 10.1016/j.commatsci.2016.10.015.